

1. Title / Name

Dr Sambasivam Gnanasekaran

2. Designation

Assistant Professor

3. Email Address

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4. Biography

Dr. Sambasivam Gnanasekaran is an Assistant Professor in the School of Computing and Data Science at Xiamen University Malaysia. He holds a Ph.D. in Computer Science and Engineering. With over a decade of comprehensive experience spanning academia, research, and industry, his expertise encompasses Artificial Intelligence, Machine Learning, and Deep Learning.

He has authored and co-authored more than 35 peer-reviewed publications, including 14 SCIE/Scopus-indexed journal articles in esteemed publishers such as Elsevier, IEEE, and Springer Nature. He is an active member of the IEEE and serves as a reviewer for several leading SCI-indexed journals and international conferences.

He currently leads a funded research project under the XMUM Research Fund (XMUMRF). In addition to his research contributions, he has held academic leadership positions, including serving as Associate Dean, where he played a key role in curriculum development and academic governance.

5. Research Interest

Artificial Intelligence, Machine Learning, Deep Learning.

6. Educational Background

- *PhD (Computer Science and Engineering), Pondicherry University, India (2018)*
- *MCA (Computer Applications), Pondicherry University, India (2009)*
- *BSc (Physics), Pondicherry University, India (2006)*

7. Working Experience

- *Assistant Professor, School of Computing and Data Science, Xiamen University Malaysia, Malaysia. (2023 - Present)*
- *Senior Assistant Professor, School of Computing Science and Engineering, VIT Bhopal University, India. (2022- 2023)*
- *Associate Professor and Associate Dean, Faculty of Information and Communication Technology, ISBAT University, Uganda (2019-2022)*
- *Assistant Professor, Department of Computer Science and Engineering, KL University, India (2017- 2019)*
- *Web Developer, Castouri Technologies, India (2009-2011)*

8. Research Experience

- *Prediction of Cassava Disease Detection using Genetic Deep Learning Convolutional Neural Network (CGDLCNN)*
Funding: XMUM Research Fund (XMUMRF) (2024 –2026). Total amount: RM59,000
- *Project Fellow in UGC Major Research Project in the Department of Computer Science, School of Engineering and Technology, Pondicherry University, India (2012-2013)*

9. Representative Publications

- *G. Sambasivam, G. Prabu kanna, Munesh Singh Chauhan, Prem Raja, Yogesh Kumar, “A Hybrid Deep Learning Model Approach for Automated Detection and Classification of Cassava Leaf Diseases”, Scientific Reports (Sci Rep) ISSN 2045-2322, (2025).*
- *G. Sambasivam, J. Amudhavel and G. Sathya, “A Predictive Performance Analysis of Vitamin D Deficiency Severity using Machine Learning Methods”, IEEE Access, 109492-109507, ISSN: 2169-3536,2020.*
- *G. Sambasivam and Geoffrey Duncan Opiyo, “A Predictive Machine Learning Application in Agriculture: Cassava Disease Detection and Classification with Imbalanced Dataset using Convolutional Neural Networks”, Egyptian Informatics Journal, Elsevier, ISSN 1110-8665.2021.*
- *Chandramohan Dhasarathan, Sambasivam Gnanasekaran, et.al, Tensor RT optimized driver drowsiness detection system using edge*

- device, *Ain Shams Engineering Journal*, Elsevier, Volume 16, Issue 10, 2025, ISSN 2090-4479.
- R.G. Babukarthik, V. Ananth Krishna Adiga, G. Sambasivam, D. Chandramohan and J. Amudhavel, "Prediction of COVID-19 using Genetic Deep Learning Convolutional Neural Network (GDCNN)", *IEEE Access*, ISSN: 2169-3536. June 2020.
 - M. Rajeswari, G. Sambasivam, T. Vengattaraman, "Appraisal and Analysis on Various Web Service Composition Approaches Based on Qos Factors", *Journal of King Saud University-Computer and Information Sciences*, Elsevier 2014.
 - L. Manjunath, N. Prabakaran, Aswin Kumer S V, E. Mohan, Balaji Natarajan, G. Sambasivam, and Vaibhav Bhushan Tyagi, "QoS Aware Integrated Management Technique for 5G mmWaveBased Hetnets", *IEEE Access*, ISSN: 2169-3536. October 2023
 - R. G. Babukarthik, D. Chandramohan, G. Sambasivam, "COVID-19 Identification in Chest X-Ray Images using Intelligent Multi-Level Classification Scenario", *Computers and Electrical Engineering*, Elsevier, ISSN: 0045-7906, 2022.
 - E. Mohan, P. Saravanan, Balaji Natarajan, Aswin Kumer S V, G. Sambasivam, G. Prabu Kanna and Vaibhav Bhushan Tyagi, "Thyroid Detection and Classification Using DNN Based on Hybrid Meta-Heuristic and LSTM Technique", *IEEE Access*, ISSN: 2169-3536. July 2023
 - Aswin Kumer S V, N Prabhakaran, E. Mohan, Balaji Natarajan, G. Sambasivam, Vaibhav Bhushan Tyagi, "Enhancing Cloud Task Scheduling with a Robust Security Approach and Optimized hybrid POA", *IEEE Access*, ISSN: 2169-3536. November 2023.
 - Moshika.M, Thirumaran, N. Balaji, Andal, G. Sambasivam & M. Rajesh, "Vulnerability Assessment in Heterogeneous Web Environment Using Probabilistic Arithmetic Automata", *IEEE Access*, ISSN: 2169-3536. May 2021
 - S. Ezhil Pradha, A. Moshika, N. Balaji, K. Andal, G. Sambasivam, M. Shanmugam, "Scheduled Access Strategy for Improving Sensor Node Battery Life Time and Delay Analysis Of Wireless Body Area Network", *IEEE Access*, ISSN: 2169-3536. Jan 2022
 - R Sivaraj Chinnasamy, Naveen J, P.J.A. Alphonse, Chandramohan Dhasarathan, G. Sambasivam, "Energy-Aware Multi-Level Clustering Scheme for Underwater Wireless Sensor Networks", *IEEE Access*, ISSN: 2169-3536. June 2022

- *Aswin Kumer S V, Lakshmi Bharath Gogu, E.Mohan , Suman Maloji , Balaji Natarajan, G. Sambasivam, Vaibhav Bhushan Tyagi, “Track and Noise Separation based on the Universal Codebook and enhanced speech recognition using Hybrid Deep Learning Method”, IEEE Access, ISSN: 2169-3536. October 2023.*